Assignment: Classification Fundamentals (Logistic Regression)
Instructions
Tasks1. Turning Regression into Classification

- Create a binary target variable:
    - Expensive (1): price above the median
    - Not Expensive (0): price at or below the median
- Briefly explain why this is now a classification problem.

2. Train a Classification Model

- Train a Logistic Regression model using:
    - Default parameters
    - A standard train/test split
- Explain in plain English:
    - What the model outputs
    - Why logistic regression is more suitable than linear regression here

3. Accuracy and Its Limitations

- Report the accuracy on the test set
- Explain why accuracy alone can be misleading for apartment prices

4. Confusion Matrix

- Compute the confusion matrix
- Describe:
    - The types of mistakes the model makes
    - Whether it misses expensive apartments or falsely labels cheap ones as expensive

5. Classification Metrics

- Compute precision and recall
- Explain what each metric means in the apartment context
- Which metric is more important for a real estate investor? Why?

6. Decision Thresholds

- Change the classification threshold (e.g., from 0.5 to 0.7 or 0.3)
- Recompute the confusion matrix and metrics
- Explain how changing the threshold affects predictions

7. Conceptual Questions (No Coding)
Answer briefly:

- What does overfitting mean in classification?
- Why can a single train/test split be misleading?
- How does cross-validation help?
- What problem does regularization solve?

Submission Guidelines

- Use clear, plain English
- Focus on understanding and interpretation
- Do not tune or optimize the model